

Optimization Lecture 5

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Outline

Convex optimization problems

Some standard convex problems

Transforming problems

Convex optimization problems

Optimization problem in standard form

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

- ▶ $x \in \mathbf{R}^n$ is the optimization variable
- ▶ $f_0 : \mathbf{R}^n \rightarrow \mathbf{R}$ is the objective or cost function
- ▶ $f_i : \mathbf{R}^n \rightarrow \mathbf{R}, i = 1, \dots, m$, are the inequality constraint functions
- ▶ $h_i : \mathbf{R}^n \rightarrow \mathbf{R}$ are the equality constraint functions

why don't consider " $<$ " and " $>$ "?

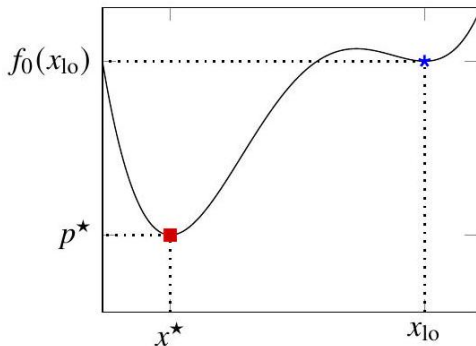
Feasible and optimal points

- ▶ $x \in \mathbf{R}^n$ is feasible if $x \in \text{dom } f_0$ and it satisfies the constraints
- ▶ optimal value is
$$p^* = \inf \{f_0(x) \mid f_i(x) \leq 0, i = 1, \dots, m, h_k(x) = 0, k = 1, \dots, p\}$$
- ▶ $p^* = \infty$ if problem is infeasible
- ▶ $p^* = -\infty$ if problem is unbounded below
- ▶ a feasible x is optimal if $f_0(x) = p^*$
- ▶ X_{opt} : the set of optimal points

Locally optimal points

x is **locally optimal** if there is an $R > 0$ such that x is optimal for

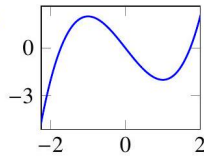
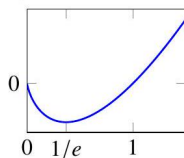
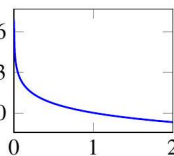
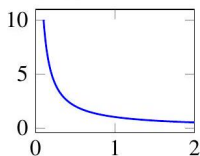
$$\begin{array}{ll}\text{minimize (over } z) & f_0(z) \\ \text{subject to} & f_i(z) \leq 0, \quad i = 1, \dots, m, \\ & h_i(z) = 0, \quad i = 1, \dots, p \\ & \|z - x\|_2 \leq R\end{array}$$



Examples

examples with $n = 1, m = p = 0$

- ▶ $f_0(x) = 1/x, \text{dom } f_0 = \mathbf{R}_{++} : p^* = 0$, no optimal point
- ▶ $f_0(x) = -\log x, \text{dom } f_0 = \mathbf{R}_{++} : p^* = -\infty$
- ▶ $f_0(x) = x \log x, \text{dom } f_0 = \mathbf{R}_{++} : p^* = -1/e, x = 1/e$ is optimal
- ▶ $f_0(x) = x^3 - 3x : p^* = -\infty, x = 1$ is locally optimal



$$f_0(x) = 1/x \quad f_0(x) = -\log x \quad f_0(x) = x \log x \quad f_0(x) = x^3 - 3x$$

Implicit and explicit constraints

standard form optimization problem has implicit constraint

$$x \in \mathcal{D} = \bigcap_{i=0}^m \text{dom } f_i \cap \bigcap_{i=1}^p \text{dom } h_i$$

- ▶ we call $\mathcal{D}$ the domain of the problem
- ▶ the constraints $f_i(x) \leq 0$, $h_i(x) = 0$ are the explicit constraints
- ▶ a problem is unconstrained if it has no explicit constraints ($m = p = 0$)

example:

$$\text{minimize } f_0(x) = - \sum_{i=1}^k \log(b_i - a_i^T x)$$

is an unconstrained problem with implicit constraints $a_i^T x < b_i$.

Feasibility problem

$$\begin{array}{ll}\text{find} & x \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

can be considered a special case of the general problem with $f_0(x) = 0$:

$$\begin{array}{ll}\text{minimize} & 0 \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

- ▶ $p^* = 0$ if constraints are feasible; any feasible x is optimal.
- ▶ $p^* = \infty$ if constraints are infeasible.

Standard form convex optimization problem

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & a_i^T x = b_i, \quad i = 1, \dots, p\end{array}$$

- ▶ objective and inequality constraints $f_0, f_1, \dots, f_m$ are convex
- ▶ equality constraints are affine, often written as $Ax = b$
- ▶ feasible and optimal sets of a convex optimization problem are convex
- ▶ problem is quasiconvex if f_0 is quasiconvex, $f_1, \dots, f_m$ are convex, $h_1, \dots, h_p$ are affine

Example

- ▶ standard form problem

$$\begin{array}{ll}\text{minimize} & f_0(x) = x_1^2 + x_2^2 \\ \text{subject to} & f_1(x) = x_1 / (1 + x_2^2) \leq 0 \\ & h_1(x) = (x_1 + x_2)^2 = 0\end{array}$$

- ▶ f_0 is convex; feasible set $\{(x_1, x_2) \mid x_1 = -x_2 \leq 0\}$ is convex
- ▶ not a convex problem (by our definition) since f_1 is not convex, h_1 is not affine
- ▶ equivalent (but not identical) to the convex problem

$$\begin{array}{ll}\text{minimize} & x_1^2 + x_2^2 \\ \text{subject to} & x_1 \leq 0 \\ & x_1 + x_2 = 0\end{array}$$

Local and global optima

Any locally optimal point of a convex problem is (globally) optimal

- ▶ suppose x is locally optimal, but there exists a feasible y with $f_0(y) < f_0(x)$
- ▶ x locally optimal means there is an $R > 0$ such that

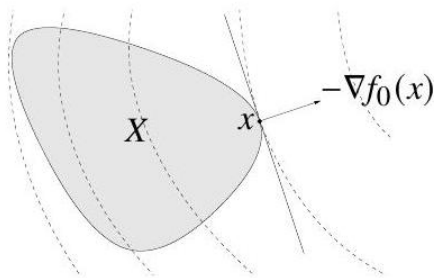
$$z \text{ feasible, } \|z - x\|_2 \leq R \implies f_0(z) \geq f_0(x)$$

- ▶ consider $z = \theta y + (1 - \theta)x$ with $\theta = R / (2\|y - x\|_2)$
- ▶ $\|y - x\|_2 > R$, so $0 < \theta < 1/2$
- ▶ z is a convex combination of two feasible points, hence also feasible
- ▶ $\|z - x\|_2 = R/2$ and $f_0(z) \leq \theta f_0(y) + (1 - \theta)f_0(x) < f_0(x)$, which contradicts our assumption that x is locally optimal

Optimality criterion for differentiable f_0

- ▶ x is optimal for a convex problem if and only if it is feasible and

$$\nabla f_0(x)^T(y - x) \geq 0 \text{ for all feasible } y$$



How to prove it?

- ▶ if nonzero, $\nabla f_0(x)$ defines a supporting hyperplane to feasible set X at x .

Examples:

- ▶ unconstrained problem (no implicit nor explicit): x minimizes $f_0(x)$ if and only if $\nabla f_0(x) = 0$
- ▶ equality constrained problem: x minimizes $f_0(x)$ subject to $Ax = b$ if and only if there exists a v such that

$$Ax = b, \quad \nabla f_0(x) + A^T v = 0$$

- ▶ minimization over nonnegative orthant: x minimizes $f_0(x)$ over $\mathbf{R}_+^n$ if and only if

$$x \geq 0, \quad \begin{cases} \nabla f_0(x)_i \geq 0 & x_i = 0 \\ \nabla f_0(x)_i = 0 & x_i > 0 \end{cases}$$

Some standard convex problems

Linear program (LP)

$$\begin{array}{ll}\text{minimize} & c^T x + d \\ \text{subject to} & Gx \leq h \\ & Ax = b\end{array}$$

- ▶ convex problem with affine objective and constraint functions
- ▶ feasible set is a polyhedron

Example: Diet problem

- ▶ Choose nonnegative quantities $x_1, \dots, x_n$ of n foods
- ▶ one unit of food j costs c_j and contains amount A_{ij} of nutrient i
- ▶ healthy diet requires nutrient i in quantity at least b_i
- ▶ to find cheapest healthy diet, solve:

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Ax \geq b, \quad x \geq 0.\end{array}$$

- ▶ express in standard LP form as

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & \begin{bmatrix} -A \\ -I \end{bmatrix} x \leq \begin{bmatrix} -b \\ 0 \end{bmatrix}\end{array}$$

Example: Piecewise-linear minimization

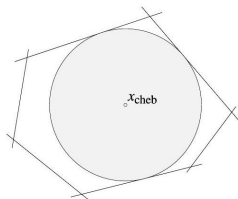
- ▶ minimize convex piecewise-linear function
$$f_0(x) = \max_{i=1,\dots,m} (a_i^T x + b_i), x \in \mathbf{R}^n$$
- ▶ equivalent to LP

$$\begin{array}{ll}\text{minimize} & t \\ \text{subject to} & a_i^T x + b_i \leq t, \quad i = 1, \dots, m\end{array}$$

with variables $x \in \mathbf{R}^n, t \in \mathbf{R}$

Example: Chebyshev center of a polyhedron

Chebyshev center of $\mathcal{P} = \{x \mid a_i^T x \leq b_i, i = 1, \dots, m\}$ is center of largest inscribed ball $\mathcal{B} = \{x_c + u \mid \|u\|_2 \leq r\}$



- $a_i^T x \leq b_i$ for all $x \in \mathcal{B}$ if and only if

$$\sup \left\{ a_i^T (x_c + u) \mid \|u\|_2 \leq r \right\} = a_i^T x_c + r \|a_i\|_2 \leq b_i$$

- x_c, r can be determined by solving LP with variables x_c, r

$$\begin{aligned} & \text{maximize} && r \\ & \text{subject to} && a_i^T x_c + r \|a_i\|_2 \leq b_i, \quad i = 1, \dots, m \end{aligned}$$

Quadratic program (QP)

$$\begin{array}{ll}\text{minimize} & (1/2)x^T Px + q^T x + r \\ \text{subject to} & Gx \leq h \\ & Ax = b\end{array}$$

- ▶ $P \in \mathbf{S}_{+}^n$, so objective is convex quadratic
- ▶ minimize a convex quadratic function over a polyhedron

Example: Least squares

- ▶ least squares problem: minimize $\|Ax - b\|_2^2$
- ▶ analytical solution $x^* = A^\dagger b$ ($A^\dagger$ is pseudo-inverse)
- ▶ can add linear constraints, e.g.,
- ▶ $x \geq 0$ (nonnegative least squares)
- ▶ $x_1 \leq x_2 \leq \dots \leq x_n$ (isotonic regression)

Example: Linear program with random cost

- ▶ LP with random cost c , with mean $\bar{c}$ and covariance Σ
- ▶ hence, LP objective $c^T x$ is random variable with mean $\bar{c}^T x$ and variance $x^T \Sigma x$
- ▶ risk-averse problem:

$$\begin{array}{ll}\text{minimize} & \mathbf{E} c^T x + \gamma \text{var}(c^T x) \\ \text{subject to} & Gx \leq h, \quad Ax = b\end{array}$$

- ▶ $\gamma > 0$ is risk aversion parameter; controls the trade-off between expected cost and variance (risk)
- ▶ express as QP

$$\begin{array}{ll}\text{minimize} & \bar{c}^T x + \gamma x^T \Sigma x \\ \text{subject to} & Gx \leq h, \quad Ax = b\end{array}$$

Quadratically constrained quadratic program (QCQP)

$$\begin{array}{ll}\text{minimize} & (1/2)x^T P_0 x + q_0^T x + r_0 \\ \text{subject to} & (1/2)x^T P_i x + q_i^T x + r_i \leq 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

- ▶ $P_i \in \mathbf{S}_{+}^n$; objective and constraints are convex quadratic
- ▶ if $P_1, \dots, P_m \in \mathbf{S}_{++}^n$, feasible region is intersection of m ellipsoids and an affine set

Second-order cone programming

$$\begin{array}{ll}\text{minimize} & f^T x \\ \text{subject to} & \|A_i x + b_i\|_2 \leq c_i^T x + d_i, \quad i = 1, \dots, m \\ & Fx = g\end{array}$$

$$(A_i \in \mathbf{R}^{n_i \times n}, F \in \mathbf{R}^{p \times n})$$

- ▶ inequalities are called second-order cone (SOC) constraints:

$$(A_i x + b_i, c_i^T x + d_i) \in \text{second-order cone in } \mathbf{R}^{n_i+1}$$

- ▶ for $n_i = 0$, reduces to an LP; if $c_i = 0$, reduces to a QCQP
- ▶ more general than QCQP and LP

Example: Robust linear programming

suppose constraint vectors a_i are uncertain in the LP

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \dots, m\end{array}$$

two common approaches to handling uncertainty

- ▶ deterministic worst-case: constraints must hold for all $a_i \in \mathcal{E}_i$ (uncertainty ellipsoids)

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i \text{ for all } a_i \in \mathcal{E}_i, \quad i = 1, \dots, m\end{array}$$

- ▶ stochastic: a_i is random variable; constraints must hold with probability η

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & \text{prob}(a_i^T x \leq b_i) \geq \eta, \quad i = 1, \dots, m\end{array}$$

Deterministic worst-case approach

- ▶ uncertainty ellipsoids are
$$\mathcal{E}_i = \{\bar{a}_i + P_i u \mid \|u\|_2 \leq 1\}, (\bar{a}_i \in \mathbf{R}^n, P_i \in \mathbf{R}^{n \times n})$$
- ▶ center of $\mathcal{E}_i$ is $\bar{a}_i$; semi-axes determined by singular values/vectors of P_i
- ▶ robust LP

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i \quad \forall a_i \in \mathcal{E}_i, \quad i = 1, \dots, m\end{array}$$

- ▶ equivalent to SOCP

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & \bar{a}_i^T x + \|P_i^T x\|_2 \leq b_i, \quad i = 1, \dots, m\end{array}$$

$$(\text{follows from } \sup_{\|u\|_2 \leq 1} (\bar{a}_i + P_i u)^T x = \bar{a}_i^T x + \|P_i^T x\|_2)$$

Stochastic approach

- ▶ assume $a_i \sim \mathcal{N}(\bar{a}_i, \Sigma_i)$
- ▶ $a_i^T x \sim \mathcal{N}(\bar{a}_i^T x, x^T \Sigma_i x)$, so

$$\text{prob} \left(a_i^T x \leq b_i \right) = \Phi \left(\frac{b_i - \bar{a}_i^T x}{\left\| \Sigma_i^{1/2} x \right\|_2} \right)$$

where $\Phi(u) = (1/\sqrt{2\pi}) \int_{-\infty}^u e^{-t^2/2} dt$ is $\mathcal{N}(0, 1)$ CDF

- ▶ **prob** $(a_i^T x \leq b_i) \geq \eta$ can be expressed as
$$\bar{a}_i^T x + \Phi^{-1}(\eta) \left\| \Sigma_i^{1/2} x \right\|_2 \leq b_i$$
- ▶ for $\eta \geq 1/2$, robust LP equivalent to SOCP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & \bar{a}_i^T x + \Phi^{-1}(\eta) \left\| \Sigma_i^{1/2} x \right\|_2 \leq b_i, \quad i = 1, \dots, m \end{array}$$

Conic form problem

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Fx + g \leq_K 0 \\ & Ax = b\end{array}$$

- ▶ constraint $Fx + g \leq_K 0$ involves a generalized inequality with respect to a proper cone K
- ▶ linear programming is a conic form problem with $K = \mathbf{R}_+^m$
- ▶ as with standard convex problem
- ▶ feasible and optimal sets are convex
- ▶ any local optimum is global

Semidefinite program (SDP)

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & x_1 F_1 + x_2 F_2 + \cdots + x_n F_n + G \leq 0 \\ & Ax = b\end{array}$$

with $F_i, G \in \mathbf{S}^k$

- ▶ inequality constraint is called linear matrix inequality (LMI)
- ▶ includes problems with multiple LMI constraints: for example,

$$x_1 \hat{F}_1 + \cdots + x_n \hat{F}_n + \hat{G} \leq 0, \quad x_1 \tilde{F}_1 + \cdots + x_n \tilde{F}_n + \tilde{G} \leq 0$$

is equivalent to single LMI

$$x_1 \begin{bmatrix} \hat{F}_1 & 0 \\ 0 & \tilde{F}_1 \end{bmatrix} + x_2 \begin{bmatrix} \hat{F}_2 & 0 \\ 0 & \tilde{F}_2 \end{bmatrix} + \cdots + x_n \begin{bmatrix} \hat{F}_n & 0 \\ 0 & \tilde{F}_n \end{bmatrix} + \begin{bmatrix} \hat{G} & 0 \\ 0 & \tilde{G} \end{bmatrix} \leq 0$$

Example: Matrix norm minimization

$$\text{minimize } \|A(x)\|_2 = \left(\lambda_{\max} \left(A(x)^T A(x) \right) \right)^{1/2}$$

where $A(x) = A_0 + x_1 A_1 + \cdots + x_n A_n$ (with given $A_i \in \mathbf{R}^{p \times q}$)
equivalent SDP

$$\begin{array}{ll} \text{minimize} & t \\ \text{subject to} & \begin{bmatrix} t\mathcal{I} & A(x) \\ A(x)^T & t\mathcal{I} \end{bmatrix} \geq 0 \end{array}$$

variables $x \in \mathbf{R}^n, t \in \mathbf{R}$

constraint follows from

$$\begin{aligned} \|A\|_2 \leq t &\iff A^T A \leq t^2 \mathcal{I}, \quad t \geq 0 \\ &\iff \begin{bmatrix} t\mathcal{I} & A \\ A^T & t\mathcal{I} \end{bmatrix} \geq 0 \end{aligned}$$

Transforming problems

Change of variables

- ▶ $\phi : \mathbf{R}^n \rightarrow \mathbf{R}^n$ is one-to-one with $\phi(\text{dom}\phi) \supseteq \mathcal{D}$
- ▶ consider (possibly non-convex) problem

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

- ▶ change variables to z with $x = \phi(z)$
- ▶ can solve equivalent problem

$$\begin{array}{ll}\text{minimize} & \tilde{f}_0(z) \\ \text{subject to} & \tilde{f}_i(z) \leq 0, \quad i = 1, \dots, m \\ & \tilde{h}_i(z) = 0, \quad i = 1, \dots, p\end{array}$$

where $\tilde{f}_i(z) = f_i(\phi(z))$ and $\tilde{h}_i(z) = h_i(\phi(z))$

- ▶ recover original optimal point as $x^* = \phi(z^*)$

Example

- ▶ non-convex problem

$$\begin{array}{ll}\text{minimize} & x_1/x_2 + x_3/x_1 \\ \text{subject to} & x_2/x_3 + x_1 \leq 1 \\ & x > 0\end{array}$$

- ▶ change variables using $x = \phi(z) = \exp z$ to get

$$\begin{array}{ll}\text{minimize} & \exp(z_1 - z_2) + \exp(z_3 - z_1) \\ \text{subject to} & \exp(z_2 - z_3) + \exp(z_1) \leq 1\end{array}$$

which is convex.

Transformation of objective and constraint functions

suppose

- ▶ ϕ_0 is monotone increasing
- ▶ $\psi_i(u) \leq 0$ if and only if $u \leq 0, i = 1, \dots, m$
- ▶ $\varphi_i(u) = 0$ if and only if $u = 0, i = 1, \dots, p$

standard form optimization problem is equivalent to

$$\begin{array}{ll}\text{minimize} & \phi_0(f_0(x)) \\ \text{subject to} & \psi_i(f_i(x)) \leq 0, \quad i = 1, \dots, m \\ & \varphi_i(h_i(x)) = 0, \quad i = 1, \dots, p\end{array}$$

example: minimizing $\|Ax - b\|$ is equivalent to minimizing $\|Ax - b\|^2$

Converting maximization to minimization

- ▶ suppose ϕ_0 is monotone decreasing
- ▶ the maximization problem

$$\begin{array}{ll}\text{maximize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

is equivalent to the minimization problem

$$\begin{array}{ll}\text{minimize} & \phi_0(f_0(x)) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

examples:

- ▶ $\phi_0(u) = -u$ transforms maximizing a concave function to minimizing a convex function
- ▶ $\phi_0(u) = 1/u$ transforms maximizing a concave positive function to minimizing a convex function

Eliminating equality constraints

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

is equivalent to

$$\begin{array}{ll}\text{minimize (over } z) & f_0(Fz + x_0) \\ \text{subject to} & f_i(Fz + x_0) \leq 0, \quad i = 1, \dots, m\end{array}$$

where F and x_0 are such that $Ax = b \iff x = Fz + x_0$ for some z

Introducing equality constraints

$$\begin{array}{ll}\text{minimize} & f_0(A_0x + b_0) \\ \text{subject to} & f_i(A_ix + b_i) \leq 0, \quad i = 1, \dots, m\end{array}$$

is equivalent to

$$\begin{array}{ll}\text{minimize (over } x, y_i) & f_0(y_0) \\ \text{subject to} & f_i(y_i) \leq 0, \quad i = 1, \dots, m \\ & y_i = A_ix + b_i, \quad i = 0, 1, \dots, m\end{array}$$

Introducing slack variables for linear inequalities

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \dots, m\end{array}$$

is equivalent to

$$\begin{array}{ll}\text{minimize(over } x, s) & f_0(x) \\ \text{subject to} & a_i^T x + s_i = b_i, \quad i = 1, \dots, m \\ & s_i \geq 0, \quad i = 1, \dots, m\end{array}$$

Epigraph form

standard form convex problem is equivalent to

$$\begin{array}{ll}\text{minimize (over } x, t) & t \\ \text{subject to} & f_0(x) - t \leq 0 \\ & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

Minimizing over some variables

$$\begin{array}{ll}\text{minimize} & f_0(x_1, x_2) \\ \text{subject to} & f_i(x_1) \leq 0, \quad i = 1, \dots, m\end{array}$$

is equivalent to

$$\begin{array}{ll}\text{minimize} & \tilde{f}_0(x_1) \\ \text{subject to} & f_i(x_1) \leq 0, \quad i = 1, \dots, m\end{array}$$

where $\tilde{f}_0(x_1) = \inf_{x_2} f_0(x_1, x_2)$